

Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device*).

The RS485 Digital IO module is sturdy, low power usage and easy to use.

16 Port DI Module: -



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators. The DIP switch for Slave ID and Baud rate are placed inside the enclosure.

The design of the modules incorporates '**resettable Fuses**' to safeguard against sudden High Voltage spikes. the board also has '**reverse polarity**' protection both for input and power connector.

the communication port also has signal isolation and additional jumper for 'end of line' 120 ohms termination resistance which may or may not be used.

Specifications

General –

I/O Connectors	2 Pin 5.08 mm pitch pluggable screw Terminal Block
Dimensions	110 mm L x 110 mm B x 50mm H
Power	Input Power – 24 V AC/DC Typical – 12V DC @ 110mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



DI Inputs –

Channels	16
Sense Voltage	4 – 40 VDC
Sense Logic	Logic '0' = <1 VDC, Logic '1' = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

Additional Features: -

Power input and communication port is isolated
Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air
EFT IEC 61000-4-4, 50A (5/50ms)
750V isolation.
CRC Error check.
No configuration needed on the IO board

Note: -

For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

DIP SETTING – For Manual Switch Setup**Configuration Settings: -**

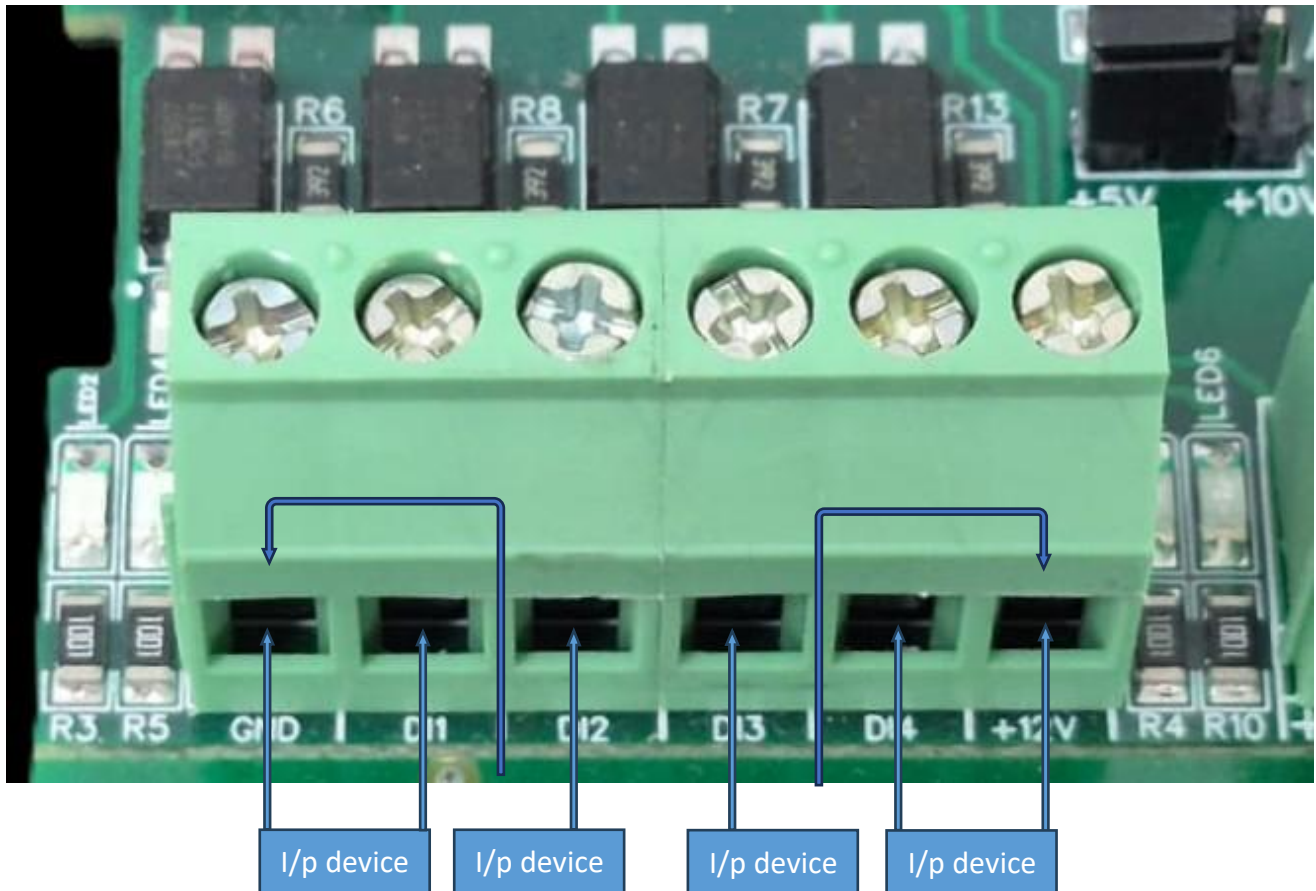
Communication Speed	9600 – 115200 Kbps (DIP SW selectable)
Data Bits	8
Parity	None
CRC	Yes
Stop bit	1
Slave ID	DIP SW selectable (1-15)
Function code DI	0x02 Read discrete Input
DI Register Address	0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15.

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded and twisted pair
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

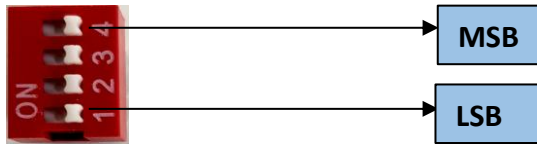
ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	DI 1(With & Without Potential)	10001	0X02	RS485	1 Bit Boolean
2	DI 2(With & Without Potential)	10002	0X02	RS485	1 Bit Boolean
3	DI 3(With & Without Potential)	10003	0X02	RS485	1 Bit Boolean
4	DI 4(With & Without Potential)	10004	0X02	RS485	1 Bit Boolean
5	DI 5(With & Without Potential)	10005	0X02	RS485	1 Bit Boolean
6	DI 6(With & Without Potential)	10006	0X02	RS485	1 Bit Boolean
7	DI 7(With & Without Potential)	10007	0X02	RS485	1 Bit Boolean
8	DI 8(With & Without Potential)	10008	0X02	RS485	1 Bit Boolean
9	DI 9(With & Without Potential)	10009	0X02	RS485	1 Bit Boolean
10	DI 10(With & Without Potential)	10010	0X02	RS485	1 Bit Boolean
11	DI 11(With & Without Potential)	10011	0X02	RS485	1 Bit Boolean
12	DI 12(With & Without Potential)	10012	0X02	RS485	1 Bit Boolean
13	DI 13(With & Without Potential)	10013	0X02	RS485	1 Bit Boolean
14	DI 14(With & Without Potential)	10014	0X02	RS485	1 Bit Boolean
15	DI 15(With & Without Potential)	10015	0X02	RS485	1 Bit Boolean
16	DI 16(With & Without Potential)	10016	0X02	RS485	1 Bit Boolean

DI



- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
 - **GND:** Common Ground reference.
 - **DI1 – DI4:** Individual Digital Input channels.
 - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
 - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
 - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

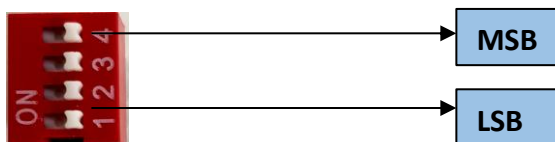
BAUD RATE DESCRIPTION



- For Baud rate Selection, DIP SW is used as per the diagram.
- Pulling up the switch will make Baud rate active.
- If no selection is made 9600 will be default Baud rate.
- When u change the Baud rate in the Module power 'ON' condition, pls press the reset button to get Change to affect.

Baud Rate	DIP SWITCH			
	1	2	3	4
9600	OFF	OFF	OFF	OFF
19200	ON	OFF	OFF	OFF
38400	OFF	ON	OFF	OFF
57600	OFF	OFF	ON	OFF
115200	OFF	OFF	OFF	ON

SLAVE ID DESCRIPTION



For Slave ID Selection SW is used to Set the SLAVE ID .

For Slave ID DIP Switch **LSB is "1"** follow through **"4" is MSB**.

Slave ID Confirmed through below Device ID table .

IF Eg. Slave ID 1 is Needed to be selected Switch number 1 should pulled up other three should be selected down side. So "1 0 0 0" will be selected as Slave ID 1.

Slave ID	DIP SWITCH				OUTPUT (Binary)	OUTPUT (Decimal)
	1	2	3	4		
0	OFF(0)	OFF(0)	OFF(0)	OFF(0)	1 0 0 0	1
1	ON(1)	OFF(0)	OFF(0)	OFF(0)	1 0 0 0	1
2	OFF(0)	ON(1)	OFF(0)	OFF(0)	0 1 0 0	2
3	ON(1)	ON(1)	OFF(0)	OFF(0)	1 1 0 0	3
4	OFF(0)	OFF(0)	ON(1)	OFF(0)	0 0 1 0	4
5	ON(1)	OFF(0)	ON(1)	OFF(0)	1 0 1 0	5
6	OFF(0)	ON(1)	ON(1)	OFF(0)	0 1 1 0	6
7	ON(1)	ON(1)	ON(1)	OFF(0)	1 1 1 0	7
8	OFF(0)	OFF(0)	OFF(0)	ON(1)	1 0 0 0	8
9	ON(1)	OFF(0)	OFF(0)	ON(1)	1 0 0 1	9
10	OFF(0)	ON(1)	OFF(0)	ON(1)	1 0 1 0	10
11	ON(1)	OFF(0)	ON(1)	ON(1)	1 0 1 1	11
12	OFF(0)	OFF(0)	ON(1)	ON(1)	1 1 0 0	12
13	ON(1)	OFF(0)	ON(1)	ON(1)	1 1 0 1	13
14	OFF(0)	ON(1)	ON(1)	ON(1)	1 1 1 0	14
15	ON(1)	ON(1)	ON(1)	ON(1)	1 1 1 1	15

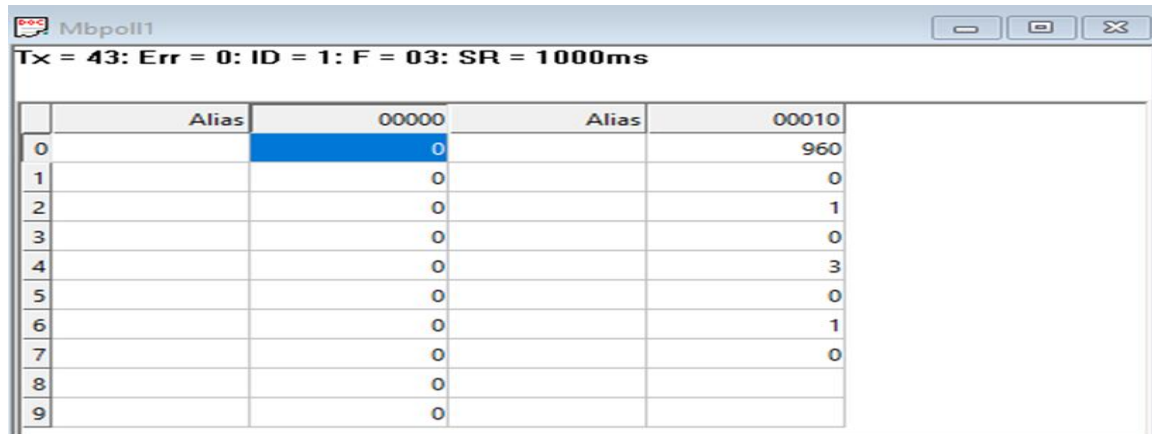
SOFT SETTING – For Modbus / Software Setup**Configuration Settings: -**

Communication Speed	9600 – 115200 (DIP SW selectable)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Soft settings (0-255).
Function code DI	0x02 Read Discrete coil
DI Register Address	0, 1, 2, 3, 4, 5, 6, 7,8,9,10,11,12,13,14,15.

Note:

1. In register 40015- you will be able to see the current parity and you can change it in register 40016
2. In register 40017- you will be able to see the current stop bit and you can change it in register 40018
3. If you want to enter the baud rate 9600/19200/38400/57600/115200etc.
Enter it as 960/1920/5760/11520etc.
4. After that press the reset button or reboot the system

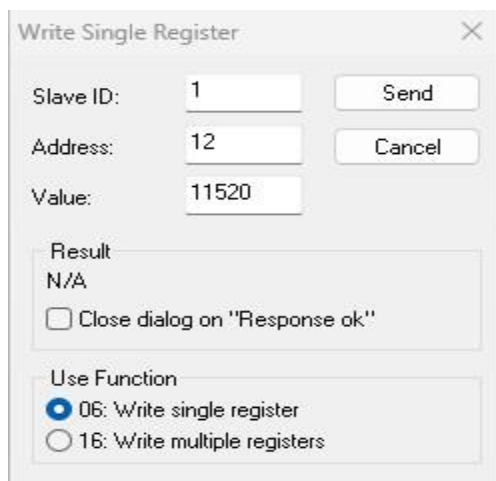
BAUD RATE and SLAVE ID DESCRIPTION:



Tx = 43: Err = 0: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		960
1		0		0
2		0		1
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

- As you can see in the image above, the resistor 40011 – 960 represents the present bund rate (9600) of the board and 40013 – 1 represents the present slave id of the board.
- To change the baud rate and slave id you must enter a value on the 40012, resistor for the baud rate and 40014 for the slave id as shown in the image below.



Write Single Register

Slave ID:

Address:

Value:

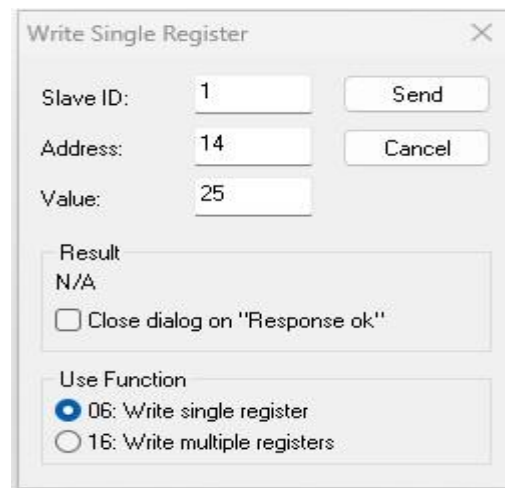
Result: N/A

☐ Close dialog on "Response ok"

Use Function:

☒ 06: Write single register

☐ 16: Write multiple registers



Write Single Register

Slave ID:

Address:

Value:

Result: N/A

☐ Close dialog on "Response ok"

Use Function:

☒ 06: Write single register

☐ 16: Write multiple registers

Mbpoll1

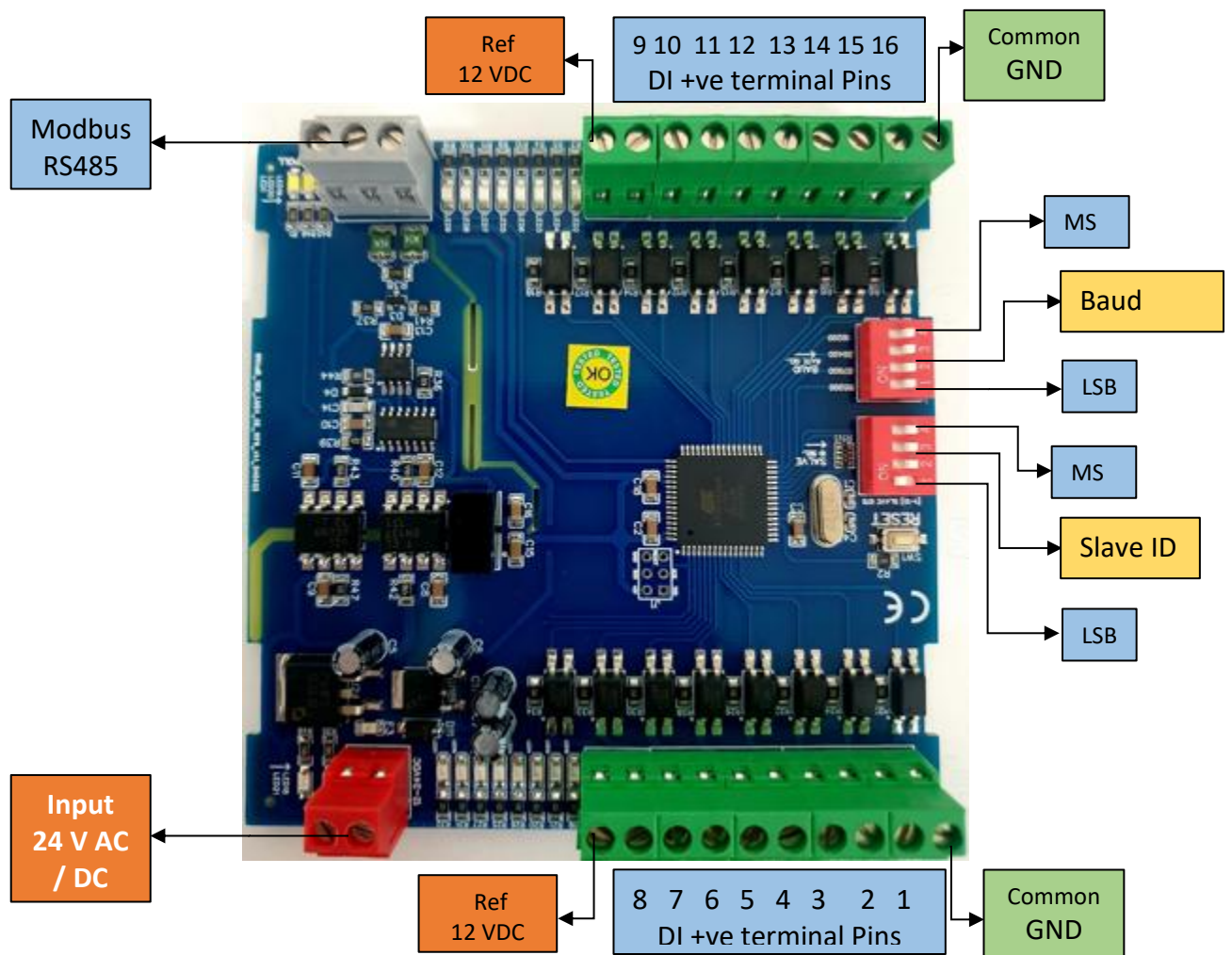
Tx = 21: Err = 0: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		960
1		0		11520
2		0		1
3		0		25
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in the image above I set 115200 baud rate and slave ID 25.

Note:

- When entering baud rate values such as 9600 / 19200 / 38400 / 57600 / 115200, enter them as 960 / 1920 / 3840 / 5760 / 11520, respectively.
- After entering the values, press the reset button or reboot the system.
- Once restarted, you will see the updated values:
- 40011 – 11520 → Baud Rate (115200)
- 40013 – 25 → Slave ID (25)
- You can also change Parity Bit and Stop Bit of the device in the same way.
- Jumper Setting (Hardware Method):
- If you place the jumper, the device will load its default communication settings (default baud rate (9600) and slave ID (1)).
- This is useful when you have many devices with different settings and want a quick way to restore defaults without software changes.
- Relation between Jumper and Reset Button:
The jumper enables the default mode, and the reset button activates it.
- Simply insert the jumper → press reset → the device restarts with factory default settings automatically



Note: <http://www.youtube.com/@wittelb>

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